

Revision — coordinates and vectors in space

[Cambridge revision]

Three numbers for a point, one product for every angle.

LESSON 61–62

2 × 40 MIN

GEOMETRY 11, CHAPTER I REVIEW

P2/P3 9.1–9.7

LESSON CLOCK

15'

25'

25'

20'

15'

Points and distances	15 min
The scalar product	25 min
Lines in space	25 min
Planes and normals	20 min
Homework	15 min

LEARNING OBJECTIVES

- Compute distances, midpoints and vectors in three dimensions.
- Use the scalar product to find an angle and to test perpendicularity.
- Write the vector equation of a line and decide whether two lines meet.
- Find the angle between a line and a plane using the normal.

TEXTBOOK

National	pp. 315–324
Cambridge	Pure Mathematics 2 & 3, pp. 190–223
Workbook	Exercise 9A–9G

01

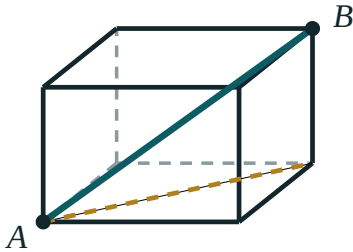
Explanation

Points and distances

Every plane formula gains a third term, and nothing else changes.

$$AB = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2 + (z_2-z_1)^2} \quad M = \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}, \frac{z_1+z_2}{2}\right)$$

Pythagoras
used twice



$$AB = \sqrt{(\Delta x^2 + \Delta y^2 + \Delta z^2)}$$

The space diagonal of a box — two applications of Pythagoras, one after the other.

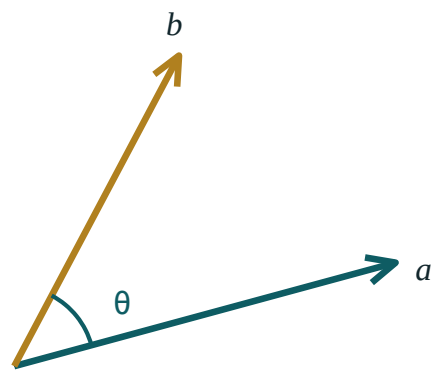
A	B	AB	M
(0,0,0)	(2,3,6)	7	(1, 1.5, 3)
(1,2,3)	(4,6,15)	13	(2.5, 4, 9)
(2,-1,4)	(6,2,8)	$\sqrt{41}$	(4, 0.5, 6)

A BOX OF EDGES a, b, c

Its space diagonal is $\sqrt{a^2 + b^2 + c^2}$ — the distance formula with the origin at one corner. Every three-dimensional distance question is this box, drawn somewhere.

The scalar product

$$u \cdot v = x_1x_2 + y_1y_2 + z_1z_2 = |u||v| \cos \theta$$



$a \cdot b = |a| \, |b| \cos \theta = a_1b_1 + a_2b_2 + a_3b_3$
perpendicular exactly when $a \cdot b = 0$

Two vectors and the angle between them — one product answers it.

$$\cos \theta = \frac{u \cdot v}{|u||v|} \quad u \perp v \Leftrightarrow u \cdot v = 0$$

<i>U</i>	<i>V</i>	<i>U</i> · <i>V</i>	θ
(1,2,2)	(2,-1,2)	4	$\approx 63.6^\circ$
(1,0,0)	(0,1,0)	0	90°
(3,4,0)	(6,8,0)	50	0°

WHY THIS REPLACES THE WHOLE OF GRADE 10’S ANGLE WORK

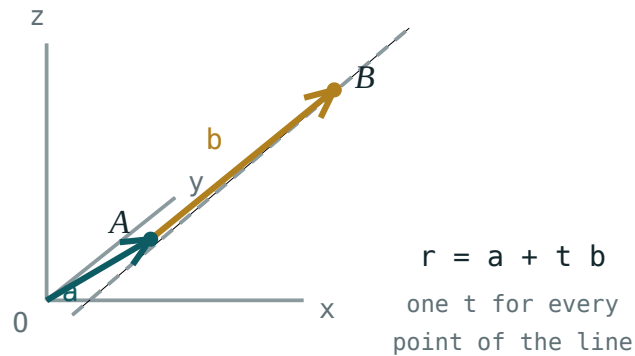
In Grade 10, an angle in space needed a projection, a right triangle and a careful drawing. Here it needs one dot product and one division. The picture is still worth drawing — but it is no longer the method.

U·*V* IS A NUMBER, NOT A VECTOR

Writing $u \cdot v = (2, 4, 4)$ confuses the scalar product with a scaling. The result of a dot product is always a single number, and its **sign** alone tells you whether the angle is acute (> 0), right ($= 0$) or obtuse (< 0).

Lines in space

$r = a + t \, d$ — through the point a , in the direction d



A point and a direction — every other point of the line is $a + t d$.

Two lines in space may be parallel, may meet, or may be **skew**. The test is a system:

1. if d_1 is a multiple of d_2 , the lines are **parallel**;
2. otherwise set $a_1 + t d_1 = a_2 + s d_2$ — three equations in t and s ;
3. solve two of them; if the third is satisfied the lines **meet**, and if not they are **skew**.

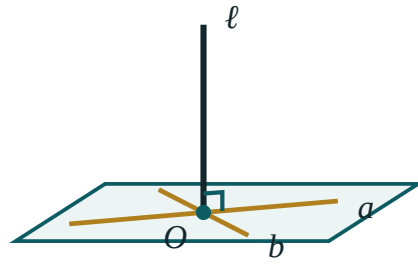
THE THIRD EQUATION IS THE WHOLE TEST

Two equations in two unknowns almost always have a solution. It is the **third** that decides, and forgetting to check it is why skew lines get reported as intersecting. Always substitute back.

The angle between two lines is the angle between their direction vectors — taken as the acute one, so use $|d_1 \cdot d_2|$.

Planes and normals

$$n \cdot r = n \cdot a, \text{ that is } ax + by + cz = d \text{ with normal } n = (a, b, c)$$



$\ell \perp a$ and $\ell \perp b$, with a and b intersecting at O
therefore ℓ is perpendicular to the whole plane

The normal — perpendicular to every line of the plane.

QUANTITY	FORMULA
angle between two planes	$\cos \theta = \frac{ \mathbf{n}_1 \cdot \mathbf{n}_2 }{ \mathbf{n}_1 \mathbf{n}_2 }$
angle between a line and a plane	$\sin \theta = \frac{ \mathbf{d} \cdot \mathbf{n} }{ \mathbf{d} \mathbf{n} }$
distance from a point to a plane	$\frac{ \mathbf{n} \cdot \mathbf{p} - d }{ \mathbf{n} }$

LINE AND PLANE USES **SINE**, NOT COSINE

The dot product gives the angle to the **normal**; the angle to the plane is its complement. So the formula carries *sin*, not *cos*. Using cosine gives the complement of the right answer — and $90^\circ - \theta$ is rarely one of the options.

Example. The angle between $\mathbf{d} = (1, 2, 2)$ and the plane $2x - y + 2z = 5$: $\mathbf{n} = (2, -1,$

$2)$, $\mathbf{d} \cdot \mathbf{n} = 2 - 2 + 4 = 4$, $|\mathbf{d}| = |\mathbf{n}| = 3$, so $\sin \theta = \frac{4}{9}$ and $\theta \approx 26.4^\circ$.

02 Worked examples

EXAMPLE 1 Find $|AB|$ and the midpoint for $A(1, 2, 3)$ and $B(4, 6, 15)$.

$$1 \quad AB = (3, 4, 12)$$

$$2 \quad |AB| = \sqrt{9 + 16 + 144} = 13$$

$$3 \quad M = (2.5, 4, 9)$$

ANSWER

13 and $(2.5, 4, 9)$

EXAMPLE 2 Find the angle between $u = (1, 2, 2)$ and $v = (2, -1, 2)$.

$$1 \quad u \cdot v = 2 - 2 + 4 = 4$$

$$2 \quad |u| = |v| = 3$$

$$3 \quad \cos \theta = \frac{4}{9}$$

$$4 \quad \theta \approx 63.6^\circ$$

ANSWER

$\approx 63.6^\circ$

EXAMPLE 3 Do the lines $r = (1, 0, 1) + t(2, 1, 1)$ and $r = (0, 1, 2) + s(1, -1, 0)$ meet?

- 1 $1 + 2t = s, t = 1 - s, 1 + t = 2$
- 2 From the third, $t = 1$; from the second, $s = 0$.
- 3 Check the first: $1 + 2 = 3 \neq 0$.
It fails.
- 4 The lines are skew.

ANSWER

Skew — the third equation is not satisfied

EXAMPLE 4 Find the angle between the line $r = a + t(1, 2, 2)$ and the plane $2x - y + 2z = 5$.

- 1 $n = (2, -1, 2)$
- 2 $d \cdot n = 2 - 2 + 4 = 4$
- 3 $\sin \theta = \frac{4}{3 \times 3} = \frac{4}{9}$
Sine, because n is the normal.
- 4 $\theta \approx 26.4^\circ$

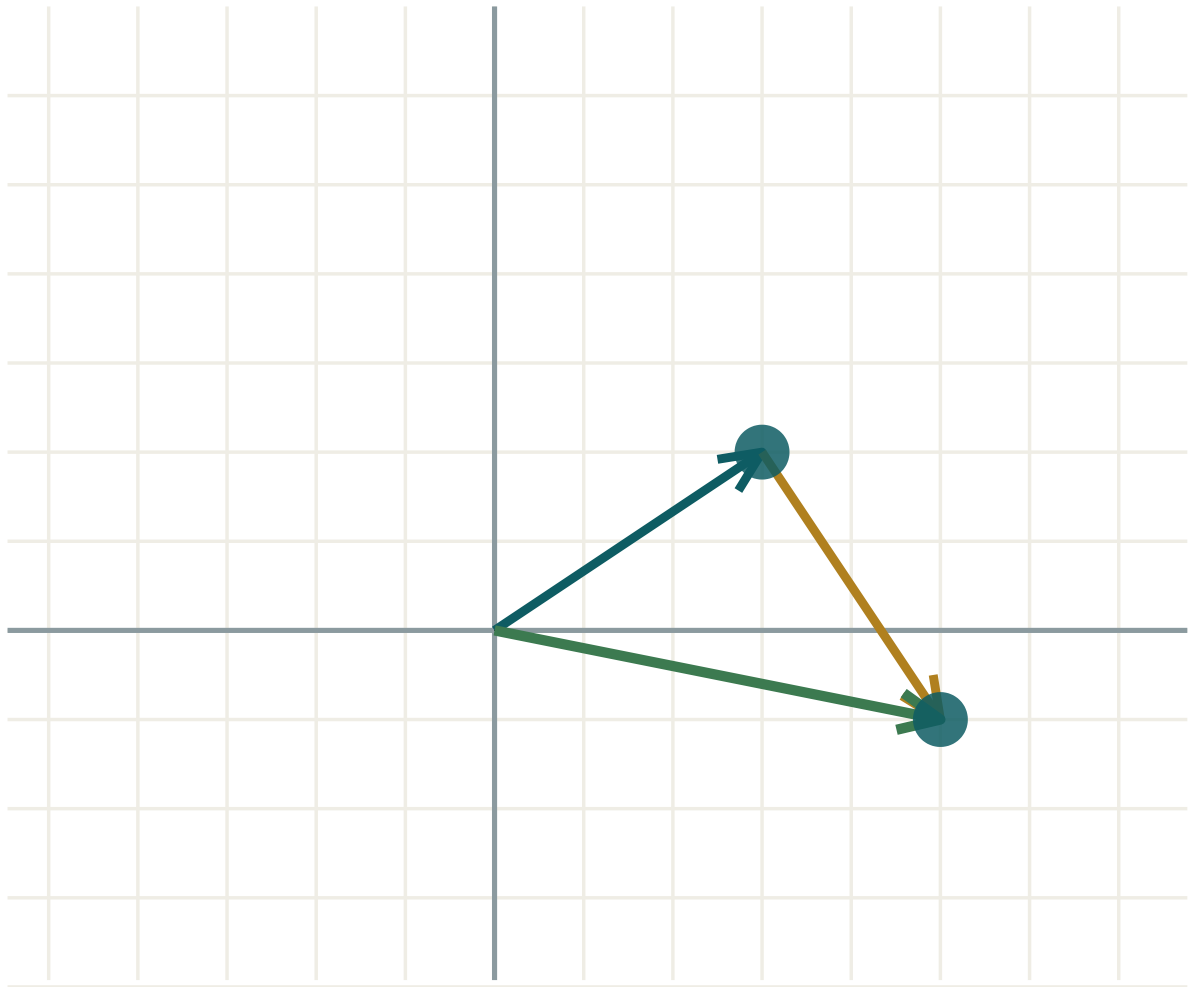
ANSWER

$\approx 26.4^\circ$

03 Interactive model

Use three metre rules taped at a corner of the room as the axes; every coordinate becomes a physical position.

Vectors and the angle between them



$$a \ (3; \ 2)$$

$$b \ (2; \ -3)$$

$$a + b \ (5; \ -1)$$

$$|a| \ 3.61$$

$$|b| \ 3.61$$

$$|a + b| \ 5.1$$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Coordinates in space	Fazoviy koordinatalar	Координаты в пространстве
Position vector	Radius-vektor	Радиус-вектор
Unit vector	Birlik vektor	Единичный вектор
Scalar (dot) product	Skalyar ko'paytma	Скалярное произведение
Perpendicular vectors	Perpendikulyar vektorlar	Перпендикулярные векторы
Vector equation of a line	To'g'ri chiziqning vektor tenglamasi	Векторное уравнение прямой
Direction vector	Yo'naltiruvchi vektor	Направляющий вектор
Normal vector	Normal vektor	Вектор нормали
Skew lines	Ayqash chiziqlar	Скрещивающиеся прямые
Parameter	Parametr	Параметр

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05

Quick check
Check yourself

$|AB|$ in space is:

$$\sqrt{\Delta x^2 + \Delta y^2}$$

$$\sqrt{\Delta x^2 + \Delta y^2 + \Delta z^2}$$

$$\Delta x + \Delta y + \Delta z$$

$$\Delta x \Delta y \Delta z$$

$u \cdot v$ is:

a vector

a number

an angle

a length

$u \perp v$ when:

$$u \cdot v = 1$$

$$u \cdot v = 0$$

$$u = v$$

$$|u| = |v|$$

A negative dot product means the angle is:

acute

right

obtuse

zero

A line in space is written:

$$y = mx + c$$

$$r = a + t d$$

$$ax + by = c$$

$$n \cdot r = d$$

Two lines are skew when:

they are parallel

two of three equations hold but not the third

they meet

they coincide

The angle between a line and a plane uses:

$\cos$

$\sin$

$\tan$

$\cot$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $|AB|$ for $A(0,0,0)$, $B(2,3,6)$

ANSWER 7

2. Midpoint of $(1,2,3)$ and $(4,6,15)$

ANSWER $(2.5, 4, 9)$

3. $(1,2,2) \cdot (2,-1,2)$

ANSWER 4

4. $|(3,4,12)|$

ANSWER 13

5. Are $(1,0,0)$ and $(0,1,0)$ perpendicular?

ANSWER Yes — dot product 0

6. Space diagonal of a $3 \times 4 \times 12$ box

ANSWER 13

7. Direction vector of $r = (1,0,1) + t(2,1,1)$

ANSWER $(2, 1, 1)$

Medium · the standard question

7 PROBLEMS

1. Angle between $(1,2,2)$ and $(2,-1,2)$

ANSWER $\approx 63.6^\circ$

2. $|AB|$ for $A(2,-1,4)$, $B(6,2,8)$

ANSWER $\sqrt{41} \approx 6.40$

3. Unit vector along $(2,3,6)$

ANSWER $\frac{1}{7}(2,3,6)$

4. Do $r = (1,0,1) + t(2,1,1)$ and $r = (0,1,2) + s(1,-1,0)$ meet?

ANSWER No — skew

5. Angle between the line $d = (1,2,2)$ and the plane $2x - y + 2z = 5$

ANSWER $\approx 26.4^\circ$

6. Normal of the plane $3x + 4y - 12z = 7$

ANSWER $(3, 4, -12)$

7. Distance from $(1, 1, 1)$ to $2x - y + 2z = 5$

ANSWER $\frac{2}{3} \approx 0.667$

Hard · stretch and reason

7 PROBLEMS

1. Angle between the planes $x + y = 1$ and $y + z = 1$

ANSWER 60°

2. Shortest distance between the two skew lines above

ANSWER $\frac{3}{\sqrt{11}} \approx 0.905$

3. A cube of edge a with a vertex at the origin: the angle between two space diagonals

ANSWER $\approx 70.5^\circ$

4. Find k so that $(1, k, 2)$ is perpendicular to $(3, -1, 1)$

ANSWER $k = 5$

5. The plane through $(1,0,0)$, $(0,1,0)$, $(0,0,1)$

ANSWER $x + y + z = 1$

6. Angle between the line $r = t(1,1,1)$ and that plane

ANSWER 90°

7. Show that $(1,2,2)$, $(2,1,-2)$ and $(2,-2,1)$ are mutually perpendicular

ANSWER All three dot products vanish

07 Homework

Homework — 5 tasks

Sketch the box for every distance question; the third coordinate is easy to lose.

- Find $|AB|$ and the midpoint for $A(-1, 3, 2)$ and $B(5, -1, 10)$.
- Find the angle between $(2, 1, 2)$ and $(1, -2, 2)$, and say whether it is acute or obtuse.
- Decide whether the lines $r = (1,1,0) + t(1,2,1)$ and $r = (2,0,1) + s(2,1,-1)$ meet, are parallel or are skew.
- Find the angle between the line with direction $(2, 2, 1)$ and the plane $x + 2y + 2z = 9$.

5. Find k so that $(3, k, -2)$ is perpendicular to $(1, 4, 5)$, and check with the dot product.